LBL – Long Baseline Underwater Localization

Main concept

- Acoustic reference nodes are deployed at known positions (seabed, surface, or water column)
- Distances are estimated using sound propagation times
- Position is estimated by trilateration

The key idea is that ranges can be estimated by measuring time differences: $r_i = c \cdot (t_\mathrm{i} - t_0)$

The name comes from the physical separation of acoustic references (as compared to a compact array on a single device)



Trilateration Geometry

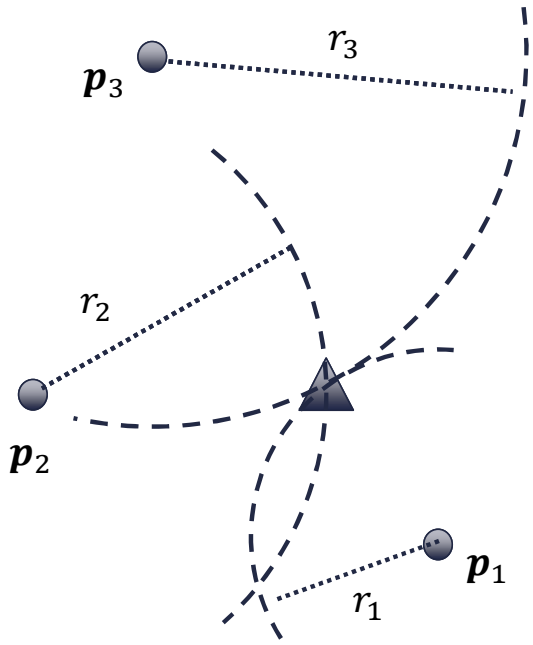
Position estimate given by the **intersection** of circumferences (spheres in 3D) with radii r_i and centers $\boldsymbol{p}_i = (x_i, y_i)$

When **multiple receivers (tracking)** are available

- 2 receivers: ambiguity
- 3 non-colinear receivers are needed to disambiguate the position in 2D (4 receivers to disambiguate in 3D)

The **inverted configuration (self-localization)** is also possible

- Beacon → receiver
- Receiver → beacon





Do we really need 3 ranges?

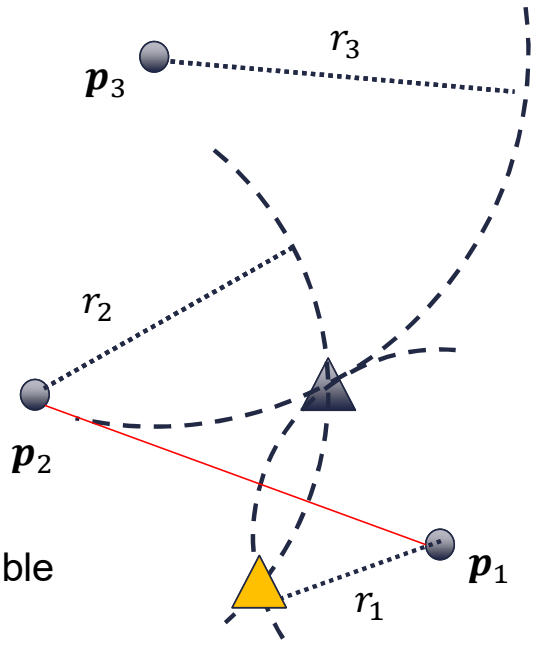
With only two ranges, the circumferences will intersect at two locations, mirrored with respect to baseline

The ambiguity disappears if we know which side of the baseline the target is

- Physical constraints
- Information from an on-line filter

In practice, three ranges (or more) are operationally preferable

- Redundancy and robustness to missing measurements
- Outlier detection
- Improved geometry
- More than 3 ranges result in an overdetermined problem, solvable e.g. by least-squares



Different Roles of Acoustic Nodes

Transmitter/Beacon - emits a known acoustic signal, possibly at known instants

Receiver - listens for signals and estimates time of arrival, possibly also decoding beacon ID/data

Transponder - receives an interrogation and replies after a known turnaround delay

Common assumptions on sound propagation:

- Constant velocity ($c \approx 1500$ m/s)
- Propagates spherically



Ranging with One-Way Travel Time (OWTT)

Simple to implement, only one sound transmission

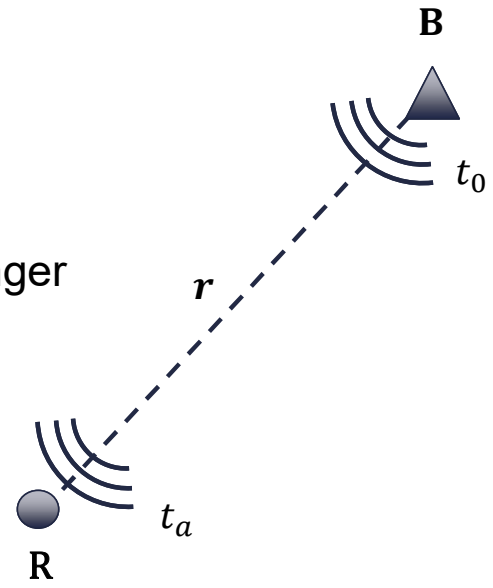
- Emission time: t_0 → may be unknown…
- Range: $r = c \cdot (t_a - t_0)$

Simplicity at receiver

- Receiver can estimate range to beacon in real time
- Receiver can work on very low power consumption
- Multiple pingers can provide additional ranges
- Multiple receivers can compute distances with same pinger

Requires synchronous clocks

- Difficult to maintain for long periods (1ms→1.5m)
- A timing error directly becomes range error
- Long-term synchronization requires atomic clocks



Ranging with Two-Way Travel Time (TWTT)

Requires emission and reception in the two nodes

- N_1 emits at $\boldsymbol{t_0}$
- N_2 receives and replies back after turnaround time $\boldsymbol{\Delta}$
- N_1 receives at $\boldsymbol{t_2 = t_0 + 2\frac{r}{c} + \Delta}$
- Range estimated at N_1: $r = \left(\frac{t_2 - t_0 - \Delta}{2}\right) \cdot c$

Avoids clock synchronization

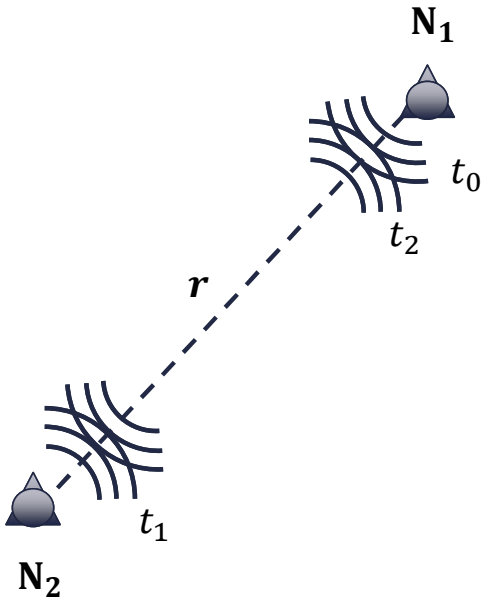
- Ideal for long term deployment

Similar devices in both ends

- Both have capability to estimate range

Difficult to scale up to a large number of nodes

- Typically one-to-one acoustic signal combinations



Main Error Sources

Acoustic Propagation

- Simples range equation assumes constant sound speed $r = c\,t$, but sound speed depends on Temperature, Salinity, and Pressure $c = c(T, S, P)$
- If sound speed varies with depth, the acoustic rays may bend, so $r \neq c_0 t$
- Transmitted signals are often affected by multipath, noise, etc.

Timing

- Clock drift/offset between nodes
- Uncertainty in latency of detection, turnaround delay, …

Geometry

- Beacon geometry affects sensitivity of trilateration solution



Moving Transmitter and/or Receiver

Relative motion of nodes has direct impact on range computation

- For a 1500m path, propagation time is ≈ 1s
- A vehicle moving at 2m/s travels 2m during that propagation time
- Impact on accuracy also depends on direction of relative motion
- Ranges with different timestamps result in inaccurate trilateration
- TWTT is more affected because if requires 2 propagation delays

Impact may be reduced with advanced processing

- Fusing LBL with other navigation information (eg. IMU, DVL) in real-time or post-processing filters (Kalman filter, particle filter, …)
- Considering each range independently into the navigation filter



Summary of LBL Strengths and Limitations

Strengths

High absolute positioning accuracy

Accuracy largely independent of surface vessel

Large operating area (~kms)

Works as an underwater GPS

Can provide redundancy with many beacons

Integrates naturally with vehicle navigation

Limitations

Requires deployment of infrastructure

Beacon positions must be surveyed accurately

Sound-speed uncertainty

Multipath / acoustic interference

OWTT needs synchronization

Geometry strongly affects accuracy

LBL localization is conceptually simple, but high-quality positioning depends as much on timing, sound propagation, beacon geometry and vehicle motion as on the trilateration equations themselves

